Alejandro Zabaleta

🚀 SOFTWARE & COMMISSIONING ENGINEER

LOGISTICS · WAREHOUSE AUTOMATION & ROBOTICS · AI

CONTACT

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LANGUAGES

Spanish	Native
English	Professional

SKILLS

LANGUAGES

- C, C++, CUDA
- Python
- Java
- JavaScript / TypeScript
- Lua, Ada, AL (Business Central)

LOGISTICS & AUTOMATION

- Intralogistics / warehouse automation
- KiSoft SRC / WCS (KNAPP)
- OSR Shuttle™ & robotic systems
- Real-time control & go-live testing
- PLC / automation fundamentals

WEB & FRAMEWORKS

- React.js, Next.js
- Node.js, Express.js
- FastAPI

CLOUD & DEVOPS

- AWS
- Docker, Kubernetes
- GitLab CI/CD
- n8n (Workflow Automation)
- Jira, Stash

DATA & AI

- SQL, MongoDB, PostgreSQL
- Pandas, NumPy
- OpenCV
- CNN, RNN, GAN

OTHER

- Data Structures & Algorithms
- Microservices Architecture
- Agile / Scrum
- Software Design Patterns

PROFILE

Software engineer currently working as a **Software Commissioning Engineer at KNAPP**, a global leader in **intralogistics and warehouse automation**. I configure and commission **KiSoft SRC** — the logistics control software that runs KNAPP's automated warehouses and robotic systems — for the **OSR Shuttle™** robotic storage-and-retrieval system and Pick-it-Easy workstations at customer distribution centers, testing material-flow processes end-to-end and driving installations to a successful go-live. Background spans AI integration, ERP development on Microsoft Dynamics 365 Business Central, and real-time C++/CUDA simulation. Open-source contributor to NASA OpenMCT and Microsoft BitNet.

EDUCATION

Bachelor Degree in Software Engineering

U-tad (2022 – 2026, completed)

Leadership Principles (Online)

Harvard University, 2024

Data Science and Machine Learning Capstone Project

IBM, Feb 2024

WORK EXPERIENCE

Software Commissioning Engineer — KiSoft SRC

KNAPP Ibérica · Warehouse Automation & Robotics | Aug 2026 – Present | Madrid, Spain · On-site

- Commission and configure **KiSoft SRC**, KNAPP's logistics control software (WCS), driving automated-warehouse and robotic installations live at customer distribution centers — including the OSR Shuttle™ robotic storage-and-retrieval system and Pick-it-Easy workstations.
- Configure, test and validate end-to-end material-flow processes on-site, coordinating testing activities through to the go-live date to hand systems over in the agreed quality and timeline.
- Troubleshoot real-time control logic across the mechanics–electrics–software stack during startup, working alongside PLC/automation layers and customer teams to stabilize and optimize throughput.
- Collaborate with international project teams, adapting the control-software configuration to each customer's operational workflows.

Software Engineer — AI & Business Central (Internship)

Arbentia | Mar 2026 – Jul 2026 | Madrid, Spain · On-site

- Built a complete Dynamics 365 Business Central extension with REST APIs, Job Queue and an integrated AI chat — learning AL from scratch and shipping a first working version in under 72 hours.
- Developed in AL and C# (Dataverse plugins) to automate business-process and financial-posting workflows on live client environments.
- Worked embedded in the functional division within the AI Business Process area, translating business requirements into technical solutions.

AI Innovation Intern

EssensUP | Sep 2025 – Nov 2025 | Remote

- Built Python data-processing pipelines to automate recurring workflows, reducing manual intervention across key operations.
- Integrated LLMs into existing production systems via REST APIs, enabling AI features in already-shipped products.
- Implemented an automatic error-tracking and analysis system, improving system-state visibility for the product team.

C++ Developer — Flight Simulator Project

Aerges | 2024 – Present · Freelance

- Developed weapon and systems-failure simulation for the F-104 Starfighter module in DCS World, covering 8+ failure scenarios across avionics, armament and hydraulic systems using C++ and Lua.
- Improved simulation modularity, enabling independent testing of failure states and reducing debugging time across a 4-person engineering team.
- Collaborated under Agile methodologies with Jira and Stash (part-time, weekends).

PERSONAL PROJECTS

- **Property Scout** — Python scraper that aggregates real estate listings from major Spanish portals (Idealista, Fotocasa) and pipes the data into the Real Estate Profitability Calculator, enabling automated portfolio analysis with live market prices. Stack: Python, BeautifulSoup, FastAPI, PostgreSQL.
- **Neural Snake** — AI that learns to play Snake using NEAT (NeuroEvolution of Augmenting Topologies), achieving 96.3% board coverage. Evolves 2,000 agents simultaneously with 64 sensory inputs, real-time training visualization and dynamic fitness shaping — built entirely from scratch without external AI frameworks. Stack: C++17, OpenGL, GLFW, ImGui, CMake.
- **Vortex Fluid Simulation** — Real-time 3D fluid simulator with dual-density SPH physics (Clavet et al. 2005), featuring a web version in JavaScript/Three.js and a native CUDA/OpenGL version with zero-copy GPU-GL interop, PBD collisions, spatial hash grid and dynamic particle scaling up to 20K particles. Stack: C++, CUDA, OpenGL, Three.js.

UNIVERSITY PROJECTS

- **Final Degree Project — Real Estate Profitability Calculator** — Full-stack web application for calculating and comparing real estate investment profitability. Covers metrics such as gross/net yield, ROI, cash flow and mortgage scenarios. Built with React and Node.js, with PostgreSQL for data persistence.
- **Autonomous Drone Follow System** — Real-time object tracking for an FPV drone using Python and OpenCV with an onboard camera feed.
- **Vehicle Recognition ML Models** — Trained and evaluated CNN, RNN and GAN architectures for vehicle-classification tasks.
- **UniTrack** — Full-stack web application built with React and Node.js that allows students and supervisors to submit, review and track academic project progress across university departments.
- **Drone Telemetry REST API** — FastAPI backend that collects, stores and exposes real-time telemetry data (position, speed, battery, tracking status) from the autonomous drone system, with MongoDB persistence and a WebSocket endpoint for live dashboard streaming.
- **Autonomous Paintball Turret** — Radar-guided turret system with Arduino control, computer-vision targeting and real-time servo actuation.